

Amazon EventBridge EC2 State Change Notification Lab

To Begin with the Lab

Summary of the Lab

In this lab, **Amazon EventBridge** was used to monitor **Amazon EC2** state-change events and send notifications to an **Amazon SNS** topic. Two EC2 instances were available as the event source, and an SNS topic with an email subscriber was created as the target. First, a broad EventBridge rule was created to capture all EC2 events, which resulted in notifications for multiple instance state transitions such as stopping, stopped, pending, and running. Then the rule was refined to filter only one specific EC2 instance and only the **Stopped** state. This showed how EventBridge rules can reduce noise and send notifications only for the events that matter. The lab ended by confirming that the email notification was triggered only when the selected instance reached the configured state.

Understanding EventBridge

Amazon EventBridge is an event bus service that receives events from AWS services and routes them to targets based on rules. It exists so you can react to AWS events without constantly polling resources or writing custom scripts. In this lab, EC2 state-change events were sent to the default event bus, and EventBridge rules decided whether the event should be forwarded to SNS. The key idea is that EventBridge does not do anything by itself unless a rule matches the incoming event pattern.

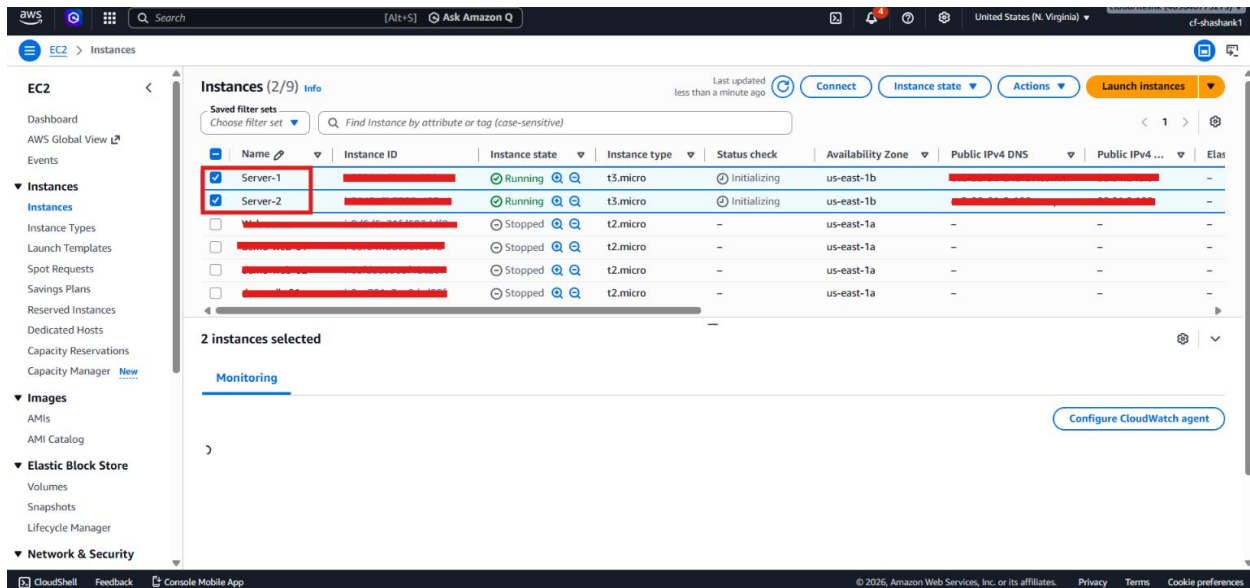
Example

If an EC2 instance changes from **running** to **stopping**, EventBridge can detect that state-change event and send an SNS notification. Later, if you only want an alert when a specific instance reaches **stopped**, you can filter the rule so that only that instance and that state trigger the message. That way, you avoid receiving emails for every minor EC2 state transition.

Lab Steps

Step 1 – Prepare the EC2 Event Source

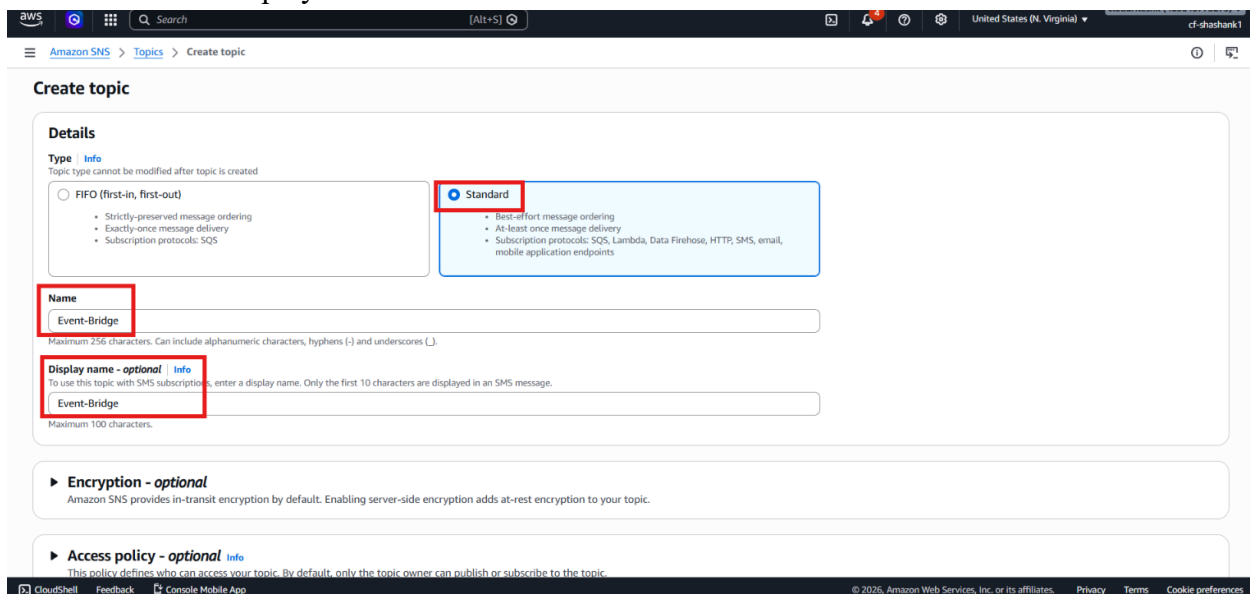
- Sign in to the AWS Management Console.
- Open **Amazon EC2**.
- Launch two small EC2 instances.



- Keep them in the default VPC.
- Use **t3.micro** or another free-tier eligible instance type.
- Keep both instances running so they can generate events.

Step 2 – Create the SNS Topic

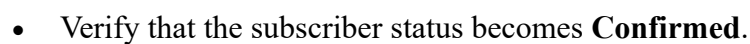
- Open Amazon SNS.
- Go to **Topics**.
- Click **Create topic**.
- Select **Standard**.
- Enter the topic name: Event-Bridge
- Set the same display name as well.



- Create the topic.

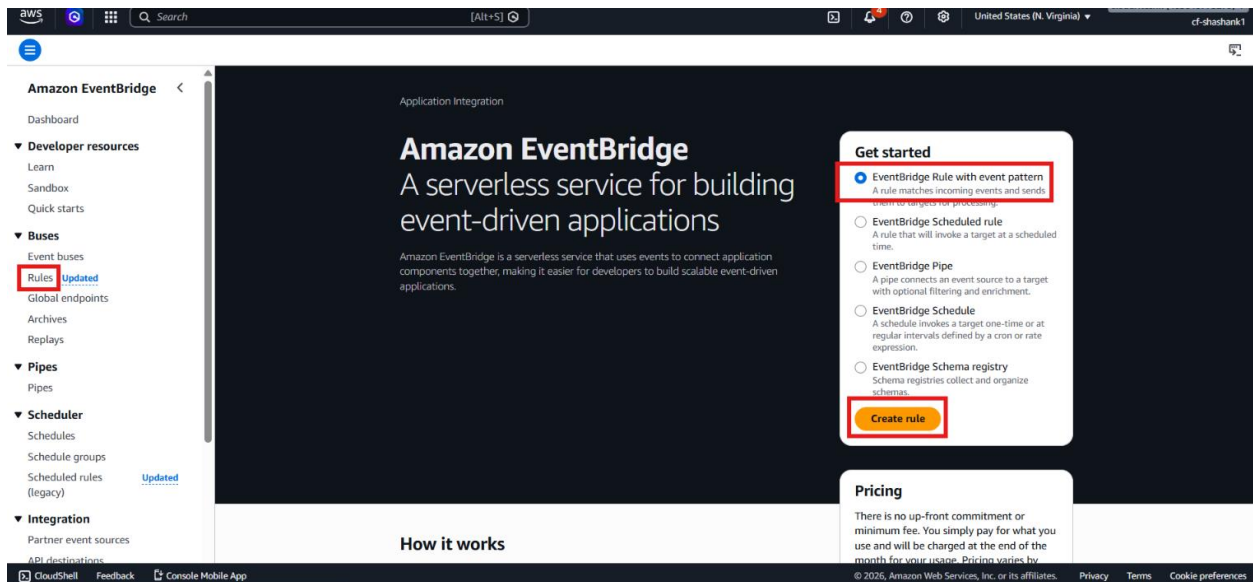
- Open the SNS topic.
- Click **Create subscription**.
- Select **Email**.
- Enter your email address as the endpoint.
- Create the subscription.

- Open the confirmation email.
- Click **Confirm subscription**.

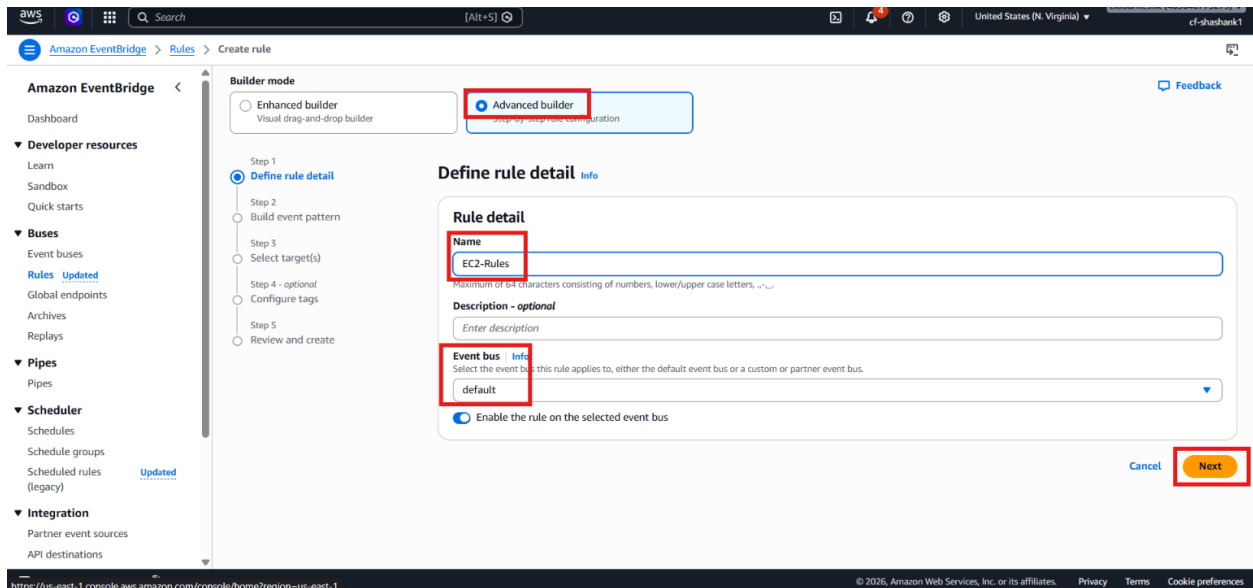


- Open **Amazon EventBridge**.

- Go to **Rules**.
- Click **Create rule**.



- Choose **Advanced builder**
- Enter the rule name: **EC2-Rules**
- Use the **Default event bus**.



- Choose the **AWS event pattern** option.
- Select **AWS services**.
- Choose **EC2** as the event source.
- Select **All events**.

Amazon EventBridge > Rules > Create rule

Step 2: build event pattern

Events
You don't have to select or enter a sample event, but it's recommended so you can reference it when writing and testing the event pattern, or filter criteria.

Event source
☒ **AWS events or EventBridge partner events**
 Events sent from AWS services or EventBridge partners.
☐ Other
 Custom events or events sent from more than one source, e.g. events from AWS services and partners.
☐ All events
 All events sent to your account.

► **Sample event - optional**
 You don't have to select or enter a sample event, but it's recommended so you can reference it when writing and testing the event pattern, or filter criteria.

Event pattern [Info](#)

Creation method
☐ Use schema
 Use an Amazon EventBridge schema to generate the event pattern.
☒ **Use pattern form**
 Use a template provided by EventBridge to create an event pattern.
☐ Custom pattern (JSON editor)
 Write an event pattern in JSON.

Event source
 AWS service or EventBridge partner as source
 AWS services

AWS service
 The name of the AWS service as the event source
 EC2

Event type
 The type of events at the source of the matching pattern
 All Events

Event pattern
 Event pattern, or filter to match the events

```
1 {
2   "source": ["aws.ec2"]
3 }
```

- Continue to the target section.
- Choose **SNS topic** as the target.
- Select the SNS topic created earlier.

Amazon EventBridge > Rules > Create rule

Step 5: Review and create

Target types
 Select an EventBridge event bus, EventBridge API destination (SaaS partner), or another AWS service as a target.
☐ EventBridge event bus
☐ EventBridge API destination
☒ **AWS service**

Select a target [Info](#)
 Select target(s) to invoke when an event matches your event pattern or when schedule is triggered (limit of 5 targets per rule)
 SNS topic

Target location
☒ **Target in this account**
☐ Target in another AWS account

Topic
 Event-Bridge

Permissions
☒ Use execution role (recommended)

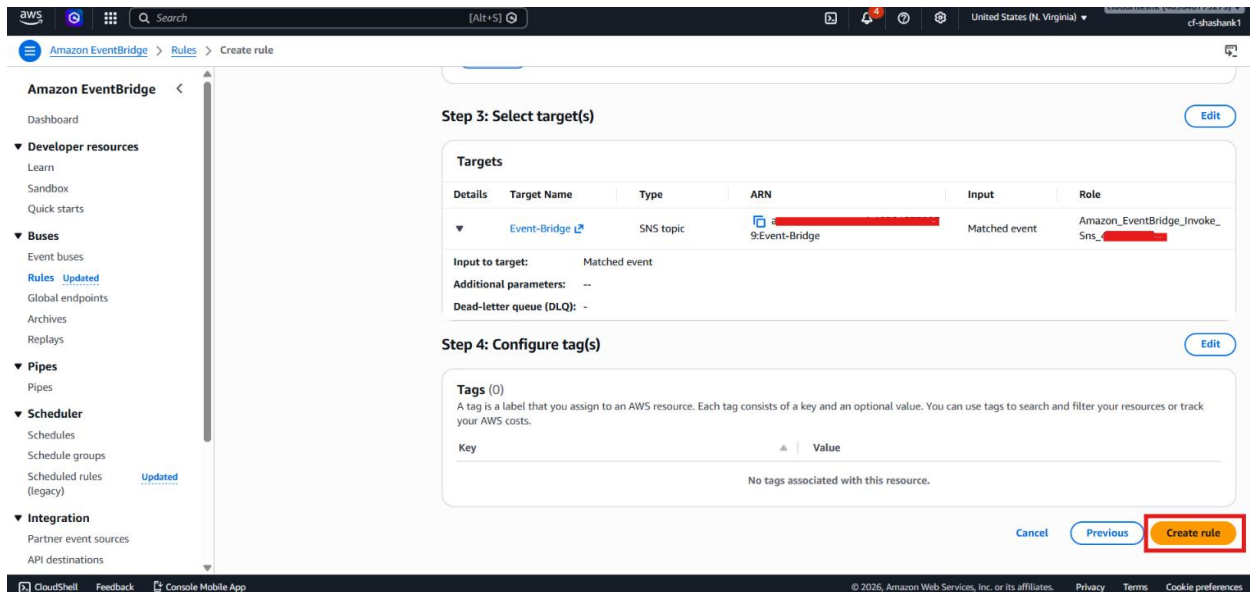
Execution role
 EventBridge needs permission to send events to the target specified above. By continuing, you are allowing us to do so. [EventBridge and AWS Identity and Access Management](#)
☒ Create a new role for this specific resource
☐ Use existing role

Role name
 Amazon_EventBridge_Invoke_Sns_437474242

► **Additional settings**

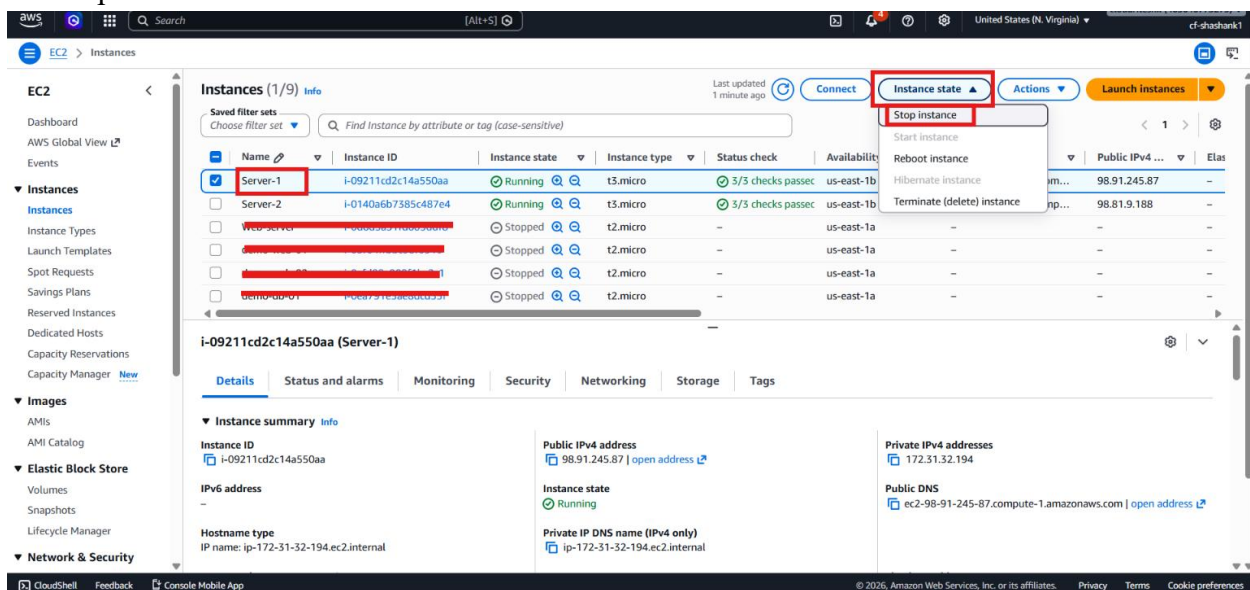
[Add another target](#) [Cancel](#) [Skip to Review and create](#) [Previous](#) [Next](#)

- Allow EventBridge to create the required IAM role automatically.
- Create the rule.

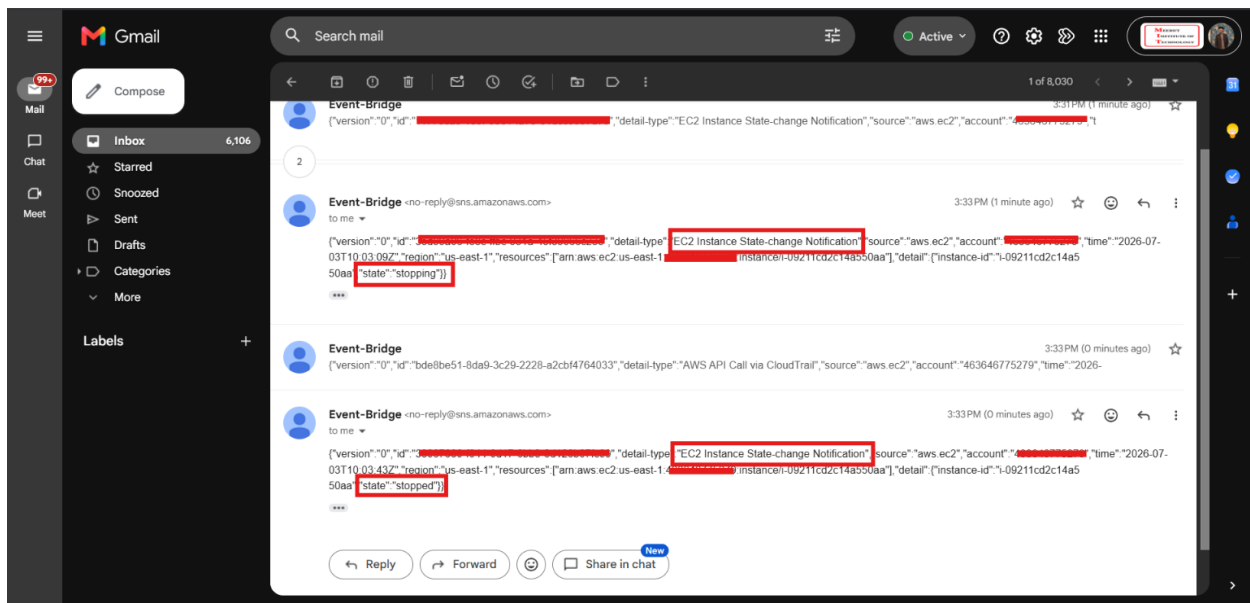


Step 5 – Test the Broad Rule

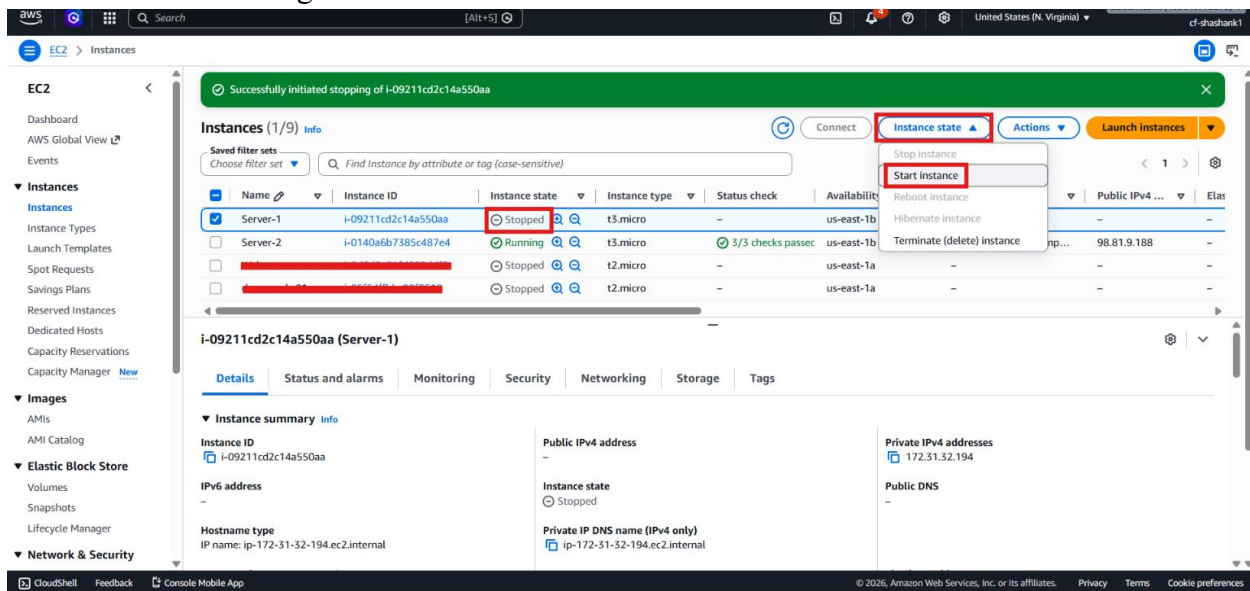
- Stop one of the EC2 instances.



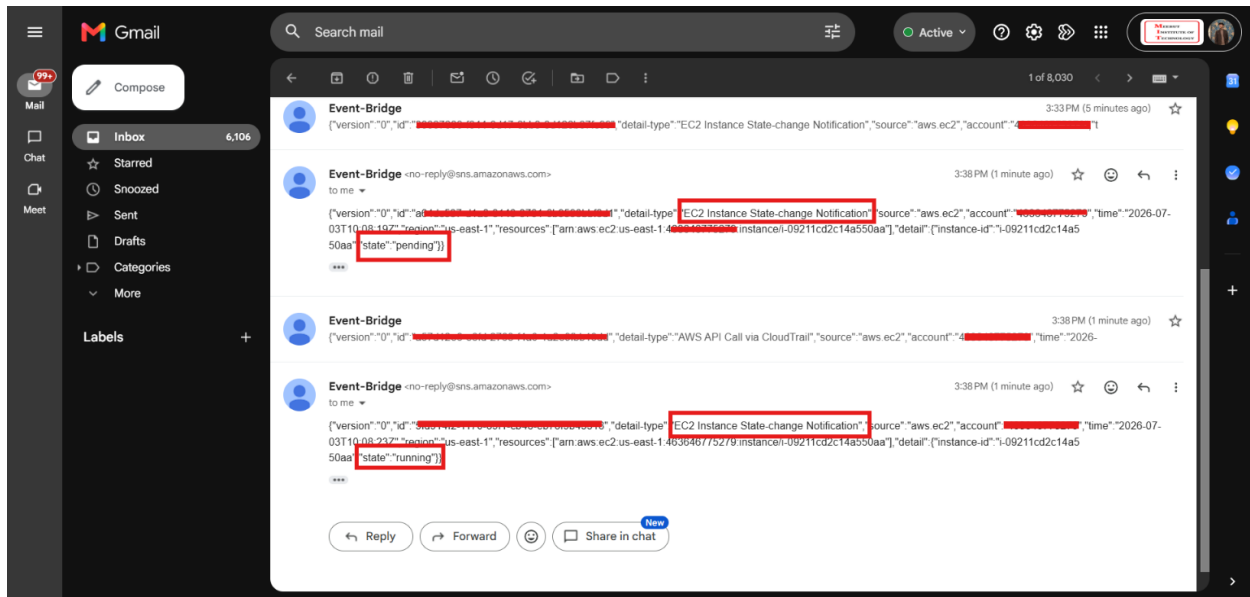
- Observe that EventBridge sends notifications for multiple state changes.



- Start the instance again.



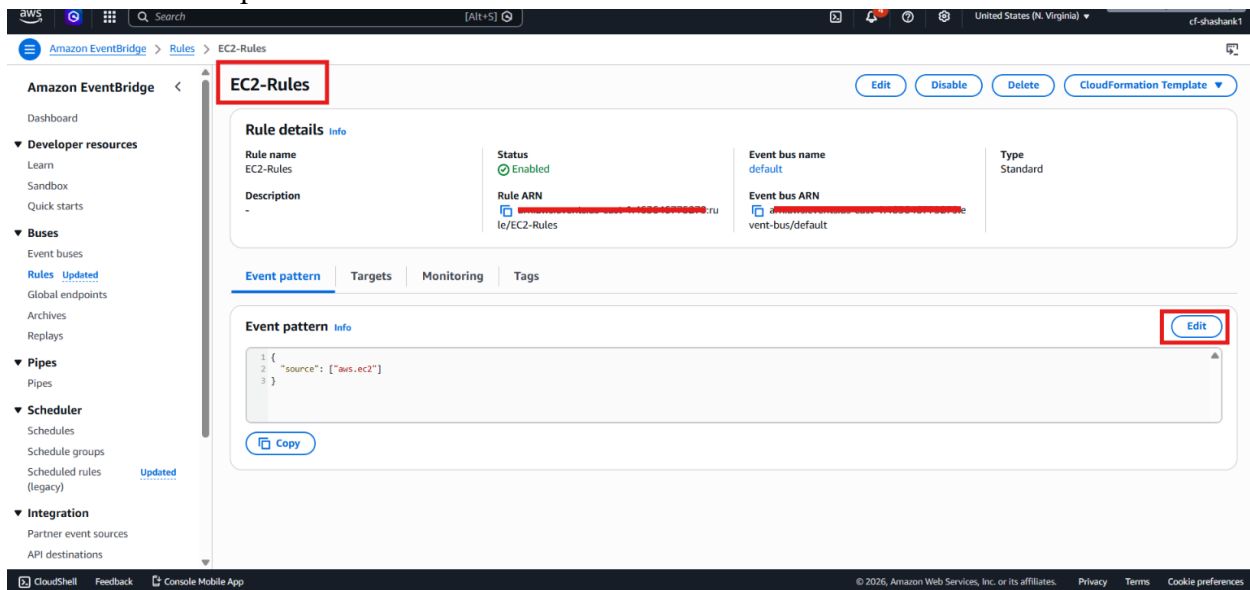
- Observe that notifications are also sent for the start sequence.



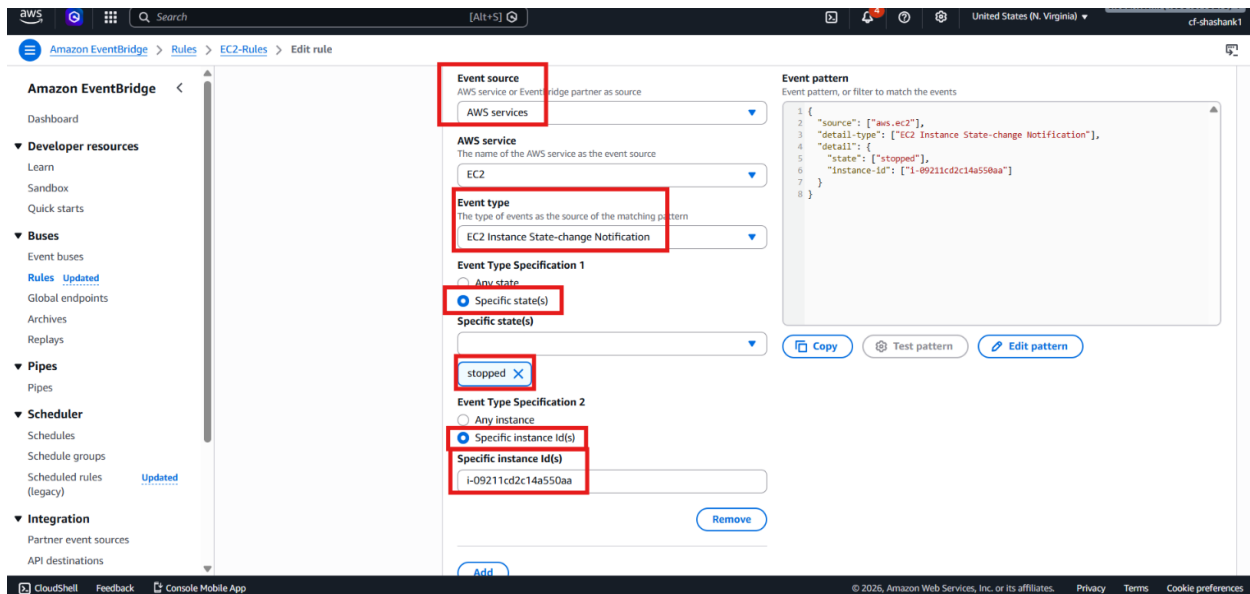
- Confirm that the rule is matching every EC2 event.

Step 6 – Refine the Rule to One Instance and One State

- Open the existing EventBridge rule.
- Edit the event pattern.



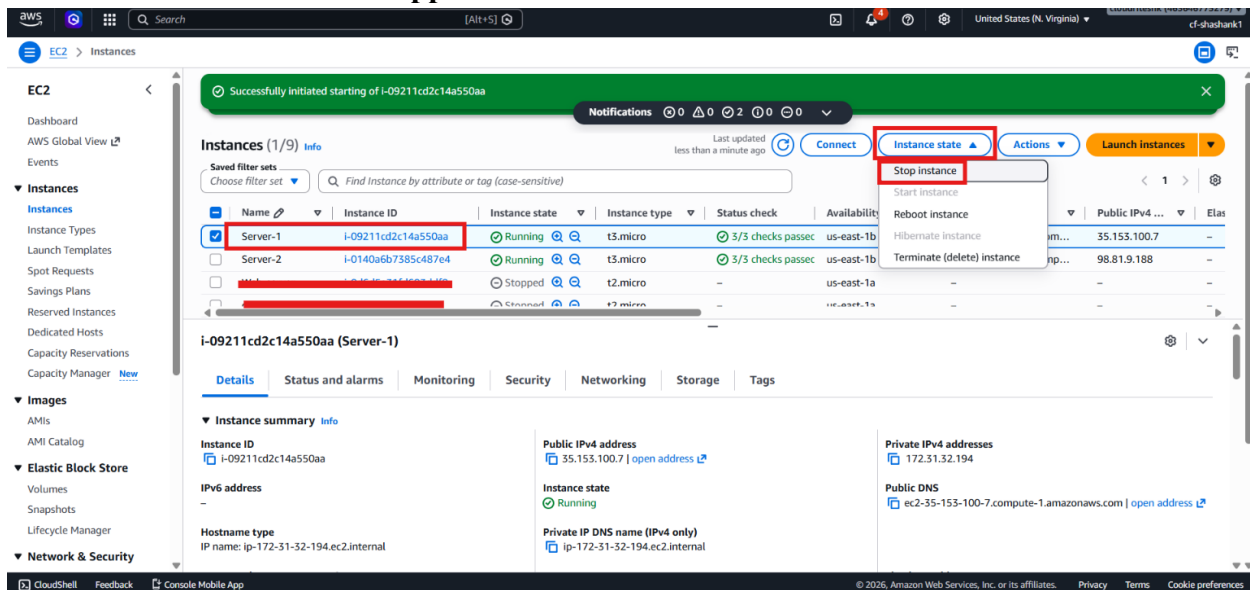
- Change the event type to **EC2 instance state-change notification**.
- Set the desired state to **Stopped**.
- Add the specific EC2 instance ID you want to monitor.



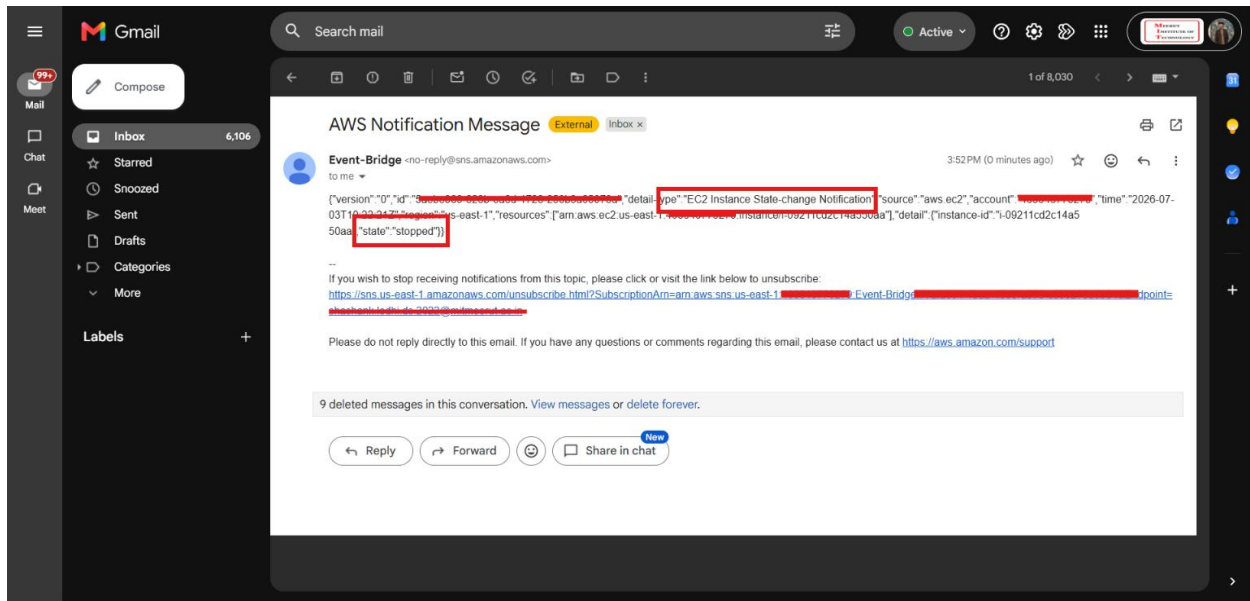
- Save the updated rule.

Step 7 – Test the Filtered Rule

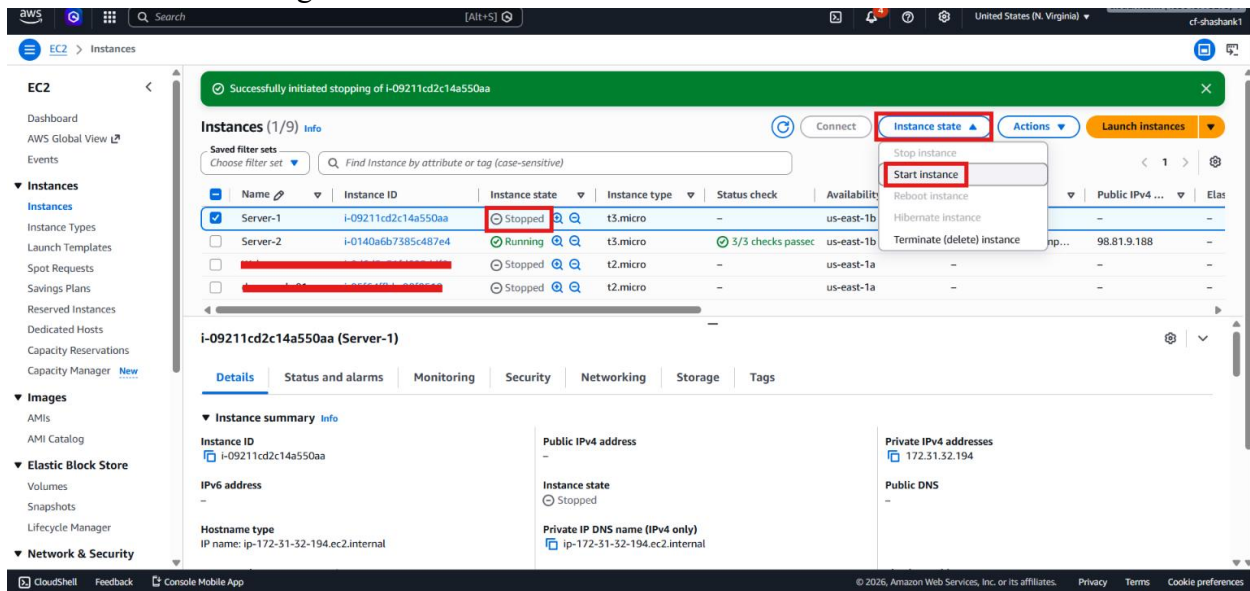
- Stop the selected EC2 instance.
- Wait until it reaches the **Stopped** state.



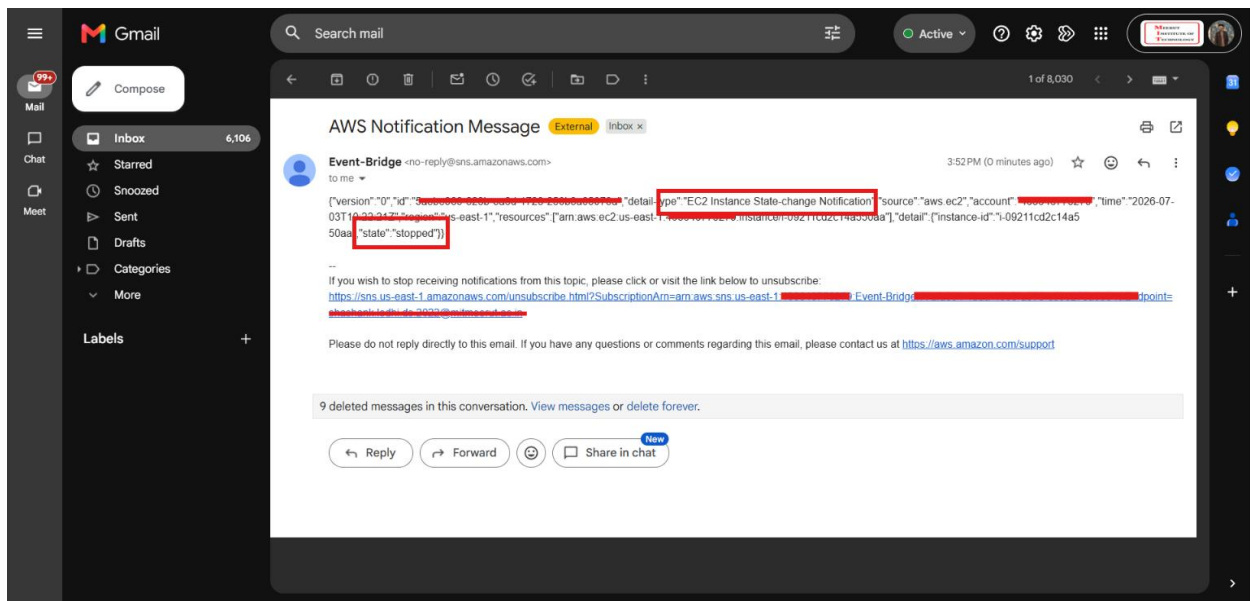
- Check your email inbox.
- Confirm that only one notification is received.



- Start the instance again.



- Confirm that no extra notification is received because the rule only matches the **Stopped** state for that specific instance.



- Stop the second EC2 instance.
- Confirm that no notification is sent because it is not part of the rule.

Verification

- Two EC2 instances were created successfully.
- An SNS topic and email subscriber were configured successfully.
- EventBridge was connected to the default event bus.
- The first rule captured all EC2 state changes.
- The rule was later refined to a single EC2 instance and the **Stopped** state.
- Email notifications were triggered only when the rule matched.
- The lab confirmed that EventBridge filtering works correctly.